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A SEMINAR REPORT
ON
IMAGE STEGANOGRAPHY

Submitted in partial fulfillment of requirements for the award of degree in

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE & ENGINEERING

Submitted By:
GOUTAM R DUBASHI
1MJ20CS072

Under the Guidance of:
Ms Sophia G
Assistant Professor, Department of Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MVJ COLLEGE OF ENGINEERING
BANGALORE-67

ACADEMIC YEAR 2023-24

MVJ COLLEGE OF ENGINEERING

Near ITPB, Whitefield, Bangalore-67

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the seminar report, entitled "**IMAGE STEGANOGRAPHY**" is a bonafide work carried out by GOUTAM R DUBASHI(1MJ20CS072) in partial fulfillment for the award of degree of 8th sem Bachelor of Engineering in Computer Science & Engineering during the academic year 2023-24. It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the Report. The seminar report has been approved as it satisfies the academic requirements.

Signature of Guide	**Signature of HOD**
Ms Sophia G	**Dr. Kiran Babu T S**

Name of examiners: Signature with date:

1.
2.

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DECLARATION

I, **GOUTAM R DUBASHI** hereby declare that the entire work titled "**IMAGE STEGANOGRAPHY"** embodied in this seminar report has been carried out by me during the 8^{th} sem of BE degree at MVJCE, Bangalore under the esteemed guidance of **Ms Sophia G** Assistant Professor, Dept of CSE, MVJCE. The work embodied in this dissertation work is original and it has not been submitted in part of full for any other degree in any University.

GOUTAM R DUBASHI
1MJ20CS072

Place:

Date:

ABSTRACT

IMAGE STEGANOGRAPHY is a technique that enables the covert transmission of information by embedding it within digital images. In today's interconnected world, where data privacy and security are paramount, steganography serves as an effective tool to ensure confidentiality by hiding sensitive data in plain sight. This abstract provides an overview of image steganography, its underlying principles, and its significance in the field of information security.

ACKNOWLEDGEMENT

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I consider it as a privilege and honor to express my sincere gratitude to our guide **Ms Sophia G , Assistant Professor,** Dept. of CSE, MVJCE, for her encouragement that has been a constant source of motivation to us for successful completion of my project.

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CHAPTER 1

INTRODUCTION

Steganography involves the seamless integration of secret data, known as the payload, into the pixels of an image without perceptible alterations to the visual appearance of the cover image. Various methods have been developed to achieve this objective, ranging from the least significant bit (**LSB**) substitution to more advanced techniques like discrete cosine transform (**DCT**) and spread spectrum modulation. These methods exploit the characteristics of digital images, such as color channels, pixel intensities, and frequency domains, to embed the payload in a manner that remains undetectable to the human eye.

The key advantages of image steganography lie in its imperceptibility and its resistance to detection. As the payload remains hidden within the image, unauthorized individuals are unlikely to even suspect the presence of concealed information. Additionally, steganographic techniques can withstand scrutiny by detection algorithms that aim to identify hidden data. This attribute makes steganography a valuable tool in various domains, including military communications, covert intelligence operations, digital watermarking, and copyright protection.

CHAPTER 2

LITERATURE SURVEY

[1] Paper Title: **"A Survey on Image Steganography Techniques and Evaluation Methods"(2020)**

Authors: **V. K. Patil, N. V. Patil, S. S. Sutar**

Summary: This survey paper provides an extensive overview of various image steganography techniques, including LSB substitution, spatial domain techniques, transform domain techniques, and adaptive methods. It also discusses different evaluation met rics used to assess the performance and security of steganographic algorithms.

[2] Paper Title: **"A Comprehensive Survey of Image Steganography Techniques" (2020)**

Authors: **R. S. Bhosale, S. B. Ghotkar**

Summary: This survey paper presents a comprehensive analysis of image steganography techniques, including both spatial and transform domain methods. It explores different embedding and extraction techniques, along with their advantages, limitations, and security implications. The paper also discusses challenges and open research areas in image steganography

[3] Paper Title: **"Recent Advances in Image Steganography: A Survey" (2018)**

Authors**: V. Thakur, D. Kapoor, A. K. Sharma**

Summary: This survey paper focuses on recent advancements in image steganography techniques. It covers topics such as evolutionary algorithms, neural networks, and deep learning-based approaches for steganography. The paper provides an in-depth analysis of each technique, their strengths, weaknesses, and potential applications.

[4] Paper Title**: "A Survey of Steganographic Techniques in Spatial and Transform Domain"(2019)**

Authors: **P. Chandramouli, D. S. Jayalakshmi**

Summary: This survey paper presents an overview of steganographic techniques in both spatial and transform domains. It discusses classic methods such as LSB substitution, as well as more advanced techniques like wavelet and DCT-based methods. The paper compares the performance, robustness, and security of these techniques and highlights their applications.

[5] Paper Title: **"A Survey on Deep Learning Techniques for Steganography and Steganalysis" (2019)**

Authors**: A. Singh, S. Sood, R. Gupta**

Summary: This survey paper focuses on the integration of deep learning techniques in image steganography and steganalysis. It reviews recent studies that leverage deep neural networks for embedding and detecting hidden information. The paper discusses various architectures, training methodologies, and feature extraction techniques used in deep learning-based steganography and steganalysis.

[6] Paper Title: **"A Survey on Secure Image Steganography Techniques" (2020)**

Authors**: P. Prashanthi, S. Vishwanath**

Summary: This survey paper provides an overview of secure image steganography techniques, emphasizing the importance of encryption in steganographic methods. It explores various encryption-based steganography techniques and discusses their effectiveness in terms of security and robustness. The paper also addresses the challenges and future research directions in secure image steganography.

CHAPTER 3

PROPOSED SYSTEM

The proposed system combines transform-domain techniques with adaptive embedding to achieve robust and secure image steganography.

1.Image Selection:

2.Transform-Domain Embedding:

3.Inverse Transform:

4.Recipient Extraction:

5.Decoding Secret Data:

Convert the extracted binary information back into the original secret message or data.
The hybrid approach of using transform-domain techniques along with adaptive embedding provides several advantages. The transform helps distribute the secret data across the image, making it more resilient to attacks and improving security. The adaptive embedding ensures that the modifications are optimized based on the cover image and secret data, enhancing both hiding capacity and visual quality.

CHAPTER 4

METHODOLOGY

The methodology for steganography involves several key steps, from selecting appropriate techniques to implementing and assessing the effectiveness of the chosen method. Here's a generalized methodology:

1. Define Objectives :

Clearly define the objectives of the steganographic communication. Determine what information needs to be concealed, who the intended recipients are, and what level of security is required.

2. Select Carrier Medium :

Choose a suitable carrier medium for hiding the secret data. Common carrier types include images, audio files, video files, and text documents. The choice depends on factors such as the nature of the information to be concealed, the intended method of transmission, and the level of scrutiny the carrier may encounter.

3. Choose Steganographic Technique:

Select an appropriate steganographic technique for embedding the secret data into the carrier medium. There are various techniques available, including:

3.1 **Least Significant Bit (LSB) insertion:** Replacing the least significant bits of pixel values in images or audio samples with the secret data.

3.2 **Spread Spectrum:** Modifying the frequency domain of a signal to hide data.

3.3 **Transform Domain Techniques:** Embedding data in the frequency or wavelet domain of a carrier signal.

3.4 **Text-Based Techniques:** Embedding data within the text of a document by subtly modifying characters, words, or formatting.

4.Embed Secret Data:

Implement the chosen steganographic technique to embed the secret data into the carrier medium. This involves encoding the secret data and modifying the carrier in a way that conceals the presence of the hidden information. Care must be taken to ensure that the modification does not significantly alter the perceptual quality of the carrier.

5.Security Measures:

Apply additional security measures as needed, such as encryption of the secret data before embedding, to protect the confidentiality and integrity of the hidden information.

6.Key Management:

If using encryption or other cryptographic techniques, carefully manage the keys used for encryption, embedding, and extraction. Secure key management is essential to prevent unauthorized access to the hidden data.

7.Testing and Evaluation:

Test the steganographic method to ensure its effectiveness and robustness. This may involve:

7.1 **Visual inspection:** Examining the carrier medium to detect any visible artifacts or anomalies introduced by the embedding process.

7.2 **Statistical analysis:** Analyzing the statistical properties of the carrier to detect deviations caused by the presence of hidden data.

7.3 **Steganalysis:** Employing steganalysis techniques to attempt to detect the presence of hidden data using automated tools or algorithms.

8.Deployment and Transmission:

Once satisfied with the effectiveness and security of the steganographic method, deploy it for actual use. Transmit the steganographic carrier to the intended recipient using appropriate communication channels, taking into account factors such as bandwidth, latency, and security considerations.

CHAPTER 5

EVALUATION

HOME PAGE:

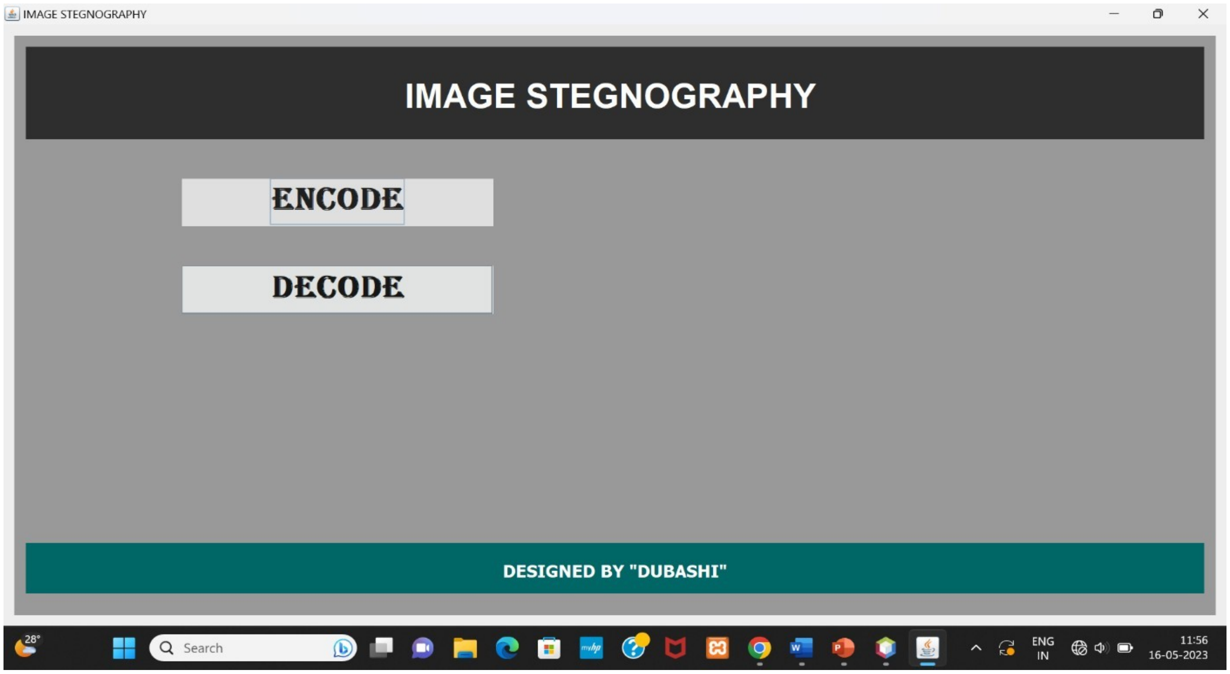

Fig 7.1: HOME PAGE

USER INTERFACE PAGE:

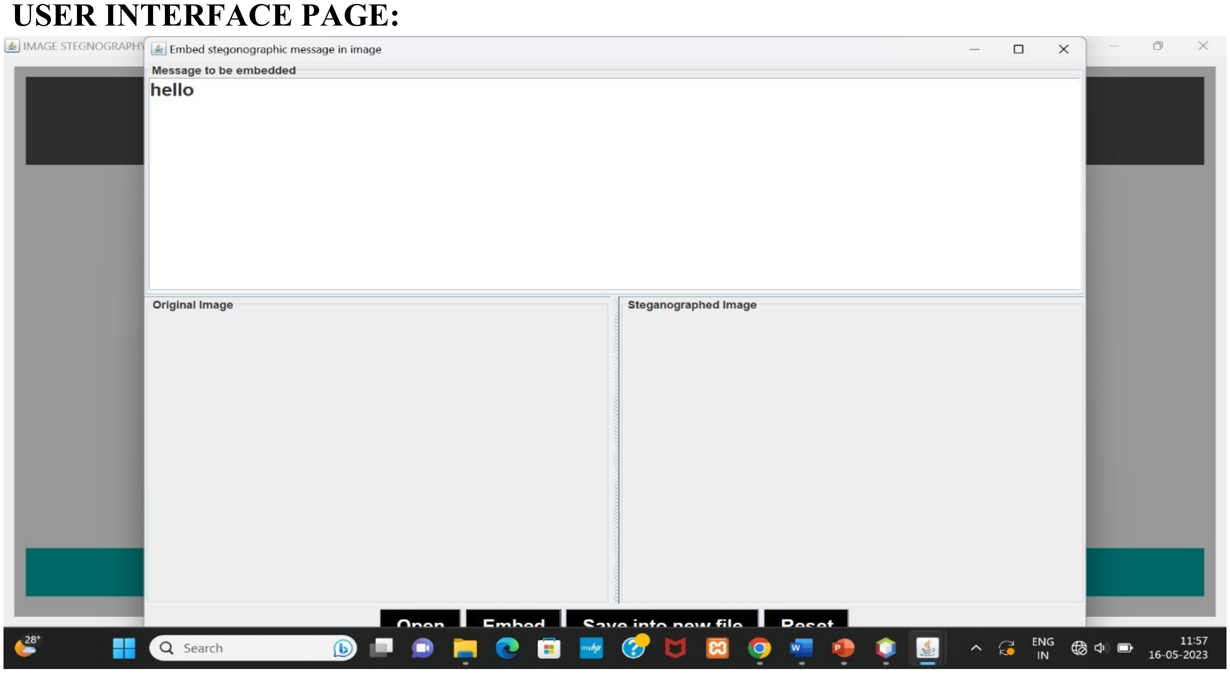

Fig 7.2: User interface Page

ENCODE DASHBOARD:

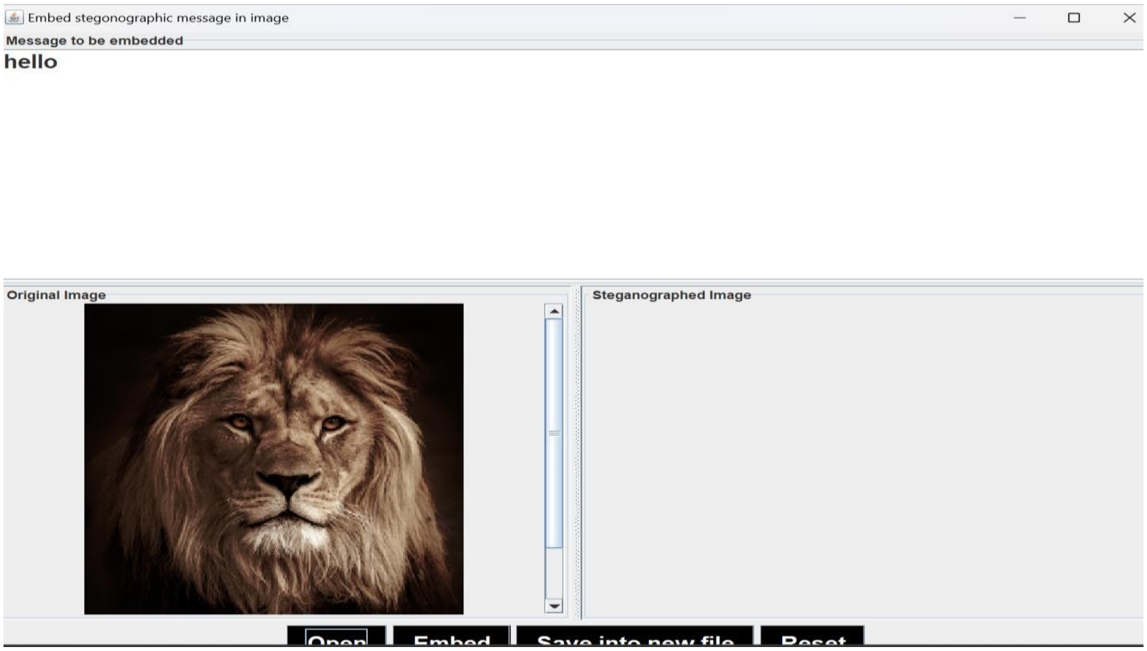

Fig 7.3: ENCODE DASHBOARD

DECODE DASHBOARD:

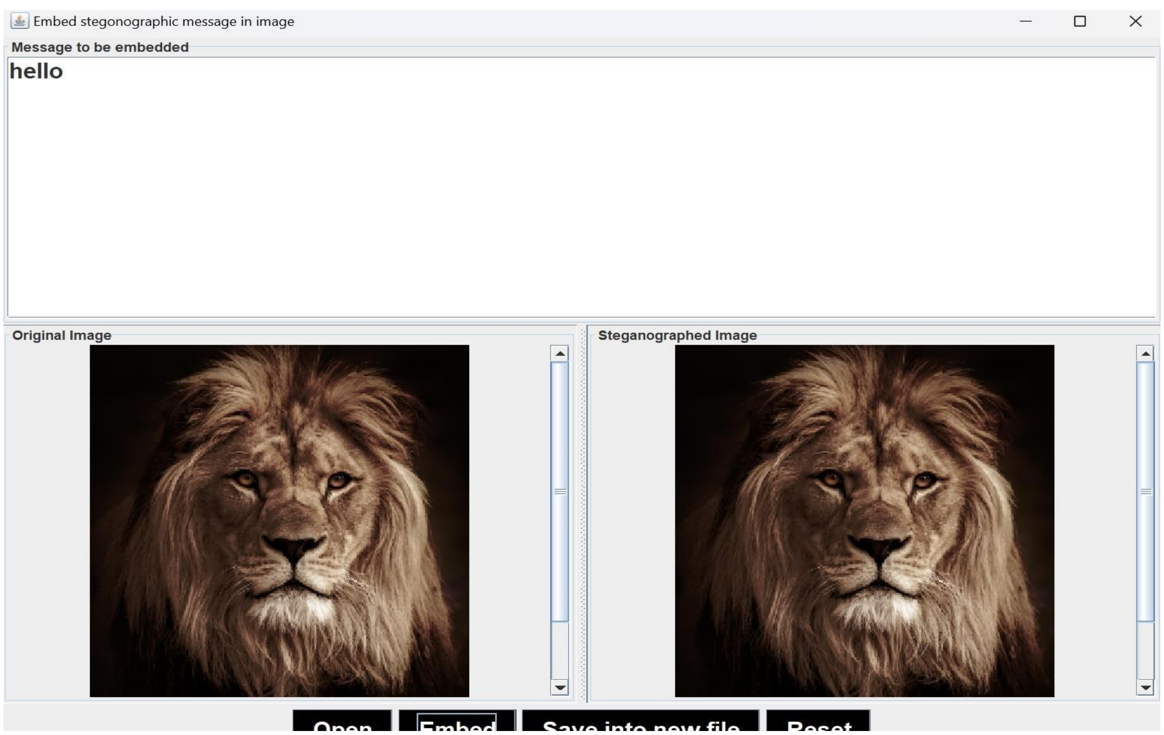

Fig 7.4: DECODE DASHBOARD

CHAPTER 6

APPLICATIONS

1. **Secure Communication:** One of the primary applications of image steganography is secure communication. By concealing sensitive information within digital images, steganography provides a covert means of transmitting confidential data without raising suspicion.

2. **Data Privacy and Confidentiality:** Image steganography can be used to protect data privacy and maintain confidentiality. Sensitive information, such as personal identification details, financial records, or medical records, can be hidden within images, ensuring that only authorized recipients can extract and access the concealed data.

3. **Digital Watermarking:** Image steganography techniques can be utilized for digital watermarking, where information is embedded within images to establish ownership or verify authenticity. Watermarks can be used to protect copyrights, prevent unauthorized distribution, or trace the source of leaked or pirated content.

4. **Covert Intelligence Operations:** Steganography plays a crucial role in covert intelligence operations, where secret information needs to be transmitted or stored securely. By hiding data within images.

5. **Digital Forensics:** Steganography plays a role in digital forensics, where investigators analyze digital media to uncover hidden information. In cases involving digital evidence, steganalysis techniques are employed to detect and extract concealed data from images.

ADVANTAGES AND DISADVANTAGES

7.1 Advantages:

1. **Cost-effective:** PHP is an open-source scripting language, meaning it is free to use and has a large community of developers contributing to its development and support. This makes PHP an affordable choice for developing a gym management system, reducing overall costs.

2. **Covert Communication:** Image steganography allows for the secret transmission of information in a covert manner. By embedding data within digital images, steganography provides a way to communicate without arousing suspicion or drawing attention to the existence of hidden information.

3. **Security through Obscurity:** Image steganography relies on the concept of security through obscurity. The hidden information remains concealed within the image, making it difficult for unauthorized individuals to even suspect its presence

4. **Resistance to Detection:** Steganographic techniques can be designed to withstand various detection methods and attacks. By carefully embedding the payload within the image, steganography can evade visual inspection and automated detection algorithms.

5. **Inconspicuous Transmission**: Images are a common and widely accepted form of media in today's digital world. By using image steganography, sensitive information can be transmitted through normal channels, such as email attachments or image sharing platforms, without raising suspicion

6. **Capacity for Large Payloads:** Digital images can provide significant capacity for hiding data. The pixels in an image offer numerous possibilities for embedding information, allowing for the concealment of large payloads.

7.2 Disadvantages:

1. **Limited Capacity:** While image steganography provides the capability to hide data within images, there is a limit to the amount of information that can be concealed without significantly degrading the image quality or raising suspicion.

2. **Susceptibility to Attacks:** While steganography aims to remain undetectable, it is not immune to attacks. Steganalysis techniques have evolved to detect steganographic content, and adversaries can employ statistical, visual, or machine learning-based methods to uncover hidden information.

3. **Fragility of Hidden Information:** Concealed data within images can be vulnerable to unintentional or intentional image modifications. Common image editing operations, such as compression, resizing, or filtering, may inadvertently alter or remove the hidden information, rendering it unrecoverable.

4. **Dependency on Image Formats:** Different image formats have distinct characteristics and compression techniques, which can affect the effectiveness and security of steganographic methods. Techniques that work well with one image format may not be as efficient or secure with another.

5. **Computational Overhead**: Embedding and extracting hidden information within images can be computationally intensive, especially when dealing with large image files or complex steganographic techniques. The process of encoding and decoding the hidden data may require significant computational resources,

CONCLUSION

In conclusion, image steganography offers a covert and secure means of transmitting information by embedding it within digital images. Its ability to maintain visual fidelity while concealing sensitive data makes it an important tool in the realm of information security. As technology advances, researchers and practitioners must continuously develop robust steganographic algorithms and detection methods to stay ahead in this ever-evolving cat-and-mouse game between concealment and detection.

FUTURE ASPECTS

1. Deep Learning-based Steganography
2. Exploring the potential of using quantum information theory and quantum computing for steganography.
3. Investigating methods to embed secret data using quantum states and quantum entanglement, providing enhanced security and capacity.
4. Investigating GAN-based models for steganalysis, enabling better detection and defense

REFERENCES

During the course of this project reference to the following materials were made

[1] A Survey on Image Steganography Techniques and Evaluation Methods,

V.K. Patil, N. V. Patil, S. S. Sutar, 2020

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